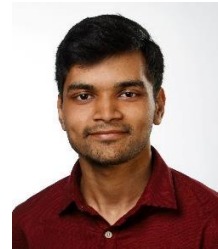


Hemant Parakh

Data Scientist · Forecasting & Applied Machine Learning

Hamburg, Germany · +49 155 66439853

hemant.parakh@tuhh.de · [linkedin.com/in/hemant-parakh](https://www.linkedin.com/in/hemant-parakh)



PROFILE

Data Science M.Sc. candidate at TU Hamburg with **two years of hands-on experience at Hapag-Lloyd AG**, where I build production forecasting models and LLM-powered analytics tools for the Yield Management organisation. I work at the intersection of time-series forecasting, commercial data and applied GenAI turning messy internal and macroeconomic data into decisions that commercial teams actually use. Available for full-time roles from **February 2027**.

PROFESSIONAL EXPERIENCE

Hapag-Lloyd AG - Working Student, Trade Analytics & Yield Management

Sep 2024 – Present

Hamburg, Germany · 20 h/week alongside full-time studies · Trade Management Analytics (2024–2026), reorganised into Yield Management (2026–present)

- Developing **freight-rate forecasting models at 7-, 14- and 30-day horizons** across global trade lanes, combining internal segmentation data (TEU, NFR, CL2) with external macroeconomic and market signals; panel models validated via walk-forward backtesting.
- Designed and shipped an **LLM-based trade-strategy pipeline** that converts a trade manager's idea into a structured volume-opportunity briefing — split research/writer prompt architecture with prompt caching and match-confidence tiering, cutting manual analysis time from 2-3 hours to 40-60 minutes.
- Built a **modular Python data-collection pipeline** for external Tier-2/3 indicators (FRED macro series, GDELT event data, container freight indices), with automated ingestion, cleaning and scheduled refresh.
- Optimised **Databricks App deployment** by migrating LLM inference to Model Serving endpoints, removing a cold-start bottleneck and reducing load time by 86%; integrated Databricks outputs into Qlik for downstream business dashboards.
- Work directly with trade managers and senior stakeholders to translate commercial questions into analytical products; including presenting concepts to non-technical directors.

Gourav Mardia & Co - Intern, Big Data & Finance

Mar 2023 – Oct 2023

Mumbai, India

- Built and maintained relational databases and data-collection systems for financial reporting workflows.
- Applied statistical analysis to client financial data and translated findings into recommendations for management.

APSIT Skills - Intern, Machine Learning

Jun 2021 – Aug 2021

Thane, India

- Built a real-time motion-detection system using background subtraction; tuned the algorithm to reduce false positives across varied lighting environments. Work published (see Publications).

EDUCATION

Technische Universität Hamburg (TUHH) - M.Sc. Data Science

Oct 2023 – Jan 2027

Focus areas: Artificial Intelligence · Big Data Analytics · Data Visualisation · Causal Data Science for Business Analytics.

Master's thesis in collaboration with Hapag-Lloyd AG(see below).

A.P. Shah Institute of Technology - B.E. Computer Engineering

2019 – 2023

Thane, India · CGPA **8.86 / 10.0** (German equivalent: **1.5**)

MASTER'S THESIS

In collaboration with Hapag-Lloyd AG - TU Hamburg in collaboration with Hapag-Lloyd AG

Jun 2026 – Dec 2026

"Data-Driven Decision Support for Forecasting and Driver Analysis in Maritime Trade"

Supervisor: Prof. Dr. Martin Kliesch, Institute for Quantum-Inspired and Quantum Optimization (E-25)

- Three research questions: identification of external macroeconomic drivers of maritime demand and freight rates, quantification of their effects, and **stability of those drivers over time** across market regimes.
- Panel modelling across trade lanes on five-plus years of weekly data, deliberately spanning multiple regimes (pre-COVID, COVID shock, peak, trough, Red Sea disruption).

- Benchmarks statistical baselines (SARIMA / SARIMAX, Prophet) against gradient-boosted models, evaluated under **walk-forward validation** rather than random splits.
- Feature engineering on internal segmentation data; lagged targets, momentum signals, CL2 mix-shift decomposition, fused with external macro and event indicators (PMI, container freight indices, Baltic Dry, GDELT).
- SHAP-based driver attribution with stability analysis across validation windows, producing interpretable output for commercial decision-makers rather than a black-box forecast.
- Research contribution: interpretable forecasting combined with driver attribution has been demonstrated in air-cargo logistics but not transferred to maritime trade, where demand dynamics and data constraints differ substantially.

INDEPENDENT RESEARCH PROJECT

Systematic Literature Review - TU Hamburg

Jun – Oct 2025

"Consumer Perception of AI-Based Software: A Systematic Literature Review on Trust and Interface Design"

Single-author review conducted to PRISMA 2020 reporting standards

- Screened **124 identified records down to 17 included reports** (13 empirical studies, 4 conceptual frameworks) through staged de-duplication, title/abstract screening and full-text assessment with logged exclusion reasons.
- Developed an a-priori two-cluster framework separating **task-specific** trust factors (objectivity, moral sensitivity, stakes) from **interface-specific** design cues (explanation format, disclosure timing, anthropomorphism, user control, privacy signals).
- Designed a five-dimension study-level certainty rubric — sample/replication, design rigour, measurement validity, transparency, statistical robustness and rated every included study Strong / Moderate / Weak with documented rationale.
- Synthesised evidence on algorithm aversion, AI disclosure timing, counterfactual explanations and the personalization-privacy trade-off into actionable design recommendations.

SELECTED PROJECTS

"Diego" - AI-powered workplace conversation simulator · AI Beavers Hackathon, Hamburg

Jun 2026

- Built the real-time voice pipeline using ElevenLabs with emotional-state modulation, so users could rehearse difficult workplace conversations against a responsive AI counterpart.

Cityblock - animated city cross-section web app

2025

- Interactive visualisation of an urban cross-section driven by live weather API data; front-end built end-to-end solo.

Student Attentiveness Detection in Classrooms

2022

- Real-time attentiveness model combining facial-expression and body-language analysis, integrated into an educational platform for actionable teaching insights.

TECHNICAL SKILLS

Languages & Tools Python (pandas, NumPy, scikit-learn, statsmodels), SQL, Git, Bash, Selenium

ML & Statistics Time-series forecasting (SARIMA/SARIMAX, Prophet), panel regression, gradient boosting, walk-forward validation, SHAP & model interpretability, feature engineering

GenAI & LLMs Prompt engineering & prompt caching, multi-agent prompt architectures, RAG, MCP servers, API integration (Anthropic, Groq, ElevenLabs)

Data Platforms Databricks (Apps, Model Serving), AWS (Lambda, core services), big-data pipelines

Research Methods PRISMA 2020 systematic review, evidence synthesis, experimental design appraisal, academic writing (LaTeX / APA)

BI & Visualisation Power BI, Tableau, Qlik, matplotlib / Plotly

Certifications AWS Cloud · Machine Learning · Relational Databases & SQL · Data Visualisation (Tableau & Power BI) · Juniper Cloud

PUBLICATIONS

- "Motion Detection using Background Subtraction Techniques", *International Engineering Journal for Research and Development*, Jan 2023.
- "Comparative Analysis of Classification Models for Indic Languages", conference paper, Oct 2022.

LANGUAGES

English C1 · Hindi C2 · German B1